

~\OneDrive\Dokumenter\AnalyseTest2After.py

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1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 # LES DATA FRA CSV FIL
6 df = pd.read_csv(r"C:\Users\danie\OneDrive\Skribbord\DatasettTest2.csv",
7                 header=None,
8                 names=["CupId", "OrderId", "CupNo", "Color", "WaterPercentage",
9                 "WeightDifference", "RealWeight", "TimeSpentMinutes"],
10                sep=";",
11                decimal=",")
12 df["TimeSpentMinutes"] = pd.to_numeric(df["TimeSpentMinutes"], errors='coerce')
13 # Convert to seconds
14 df["TimeSpentSeconds"] = df["TimeSpentMinutes"] * 60
15
16 WEIGHT_AT_100_PERCENT_G = 28.62 #29.22 på test 1 pga. tareringsfeil
17 CUP_WEIGHT_G = 14.61 #15.21 på test 1 pga. tareringsfeil
18 NET_WEIGHT_AT_100_PERCENT_G = WEIGHT_AT_100_PERCENT_G - CUP_WEIGHT_G # = 13g
19
20 df["TargetWeight"] = CUP_WEIGHT_G + (df["WaterPercentage"] / 100) * NET_WEIGHT_AT_100_PERCENT_G
21 df["WeightDifference"] = df["RealWeight"] - df["TargetWeight"]
22
23 print(f"Manglende verdier:\n{df.isnull().sum()}")
24 # OUTLIERS PÅ WeightDifference
25 abs_diff = df['WeightDifference'].abs()
26 Q1 = abs_diff.quantile(0.25)
27 Q3 = abs_diff.quantile(0.75)
28 IQR = Q3 - Q1
29 outliers_weight = df[abs_diff > Q3 + 1.5 * IQR]
30
31 # OUTLIERS PÅ TimeSpentMinutes
32 Q1 = df['TimeSpentMinutes'].quantile(0.25)
33 Q3 = df['TimeSpentMinutes'].quantile(0.75)
34 IQR = Q3 - Q1
35 outliers_time = df[(df['TimeSpentMinutes'] < Q1 - 1.5 * IQR) | (df['TimeSpentMinutes'] > Q3 + 1.5 * IQR)]
36
37 outliers = pd.concat([outliers_weight, outliers_time]).drop_duplicates()
38 df_with_outliers_removed = df.drop(outliers.index)
39 print("\n" + f"Antall outliers funnet: {len(outliers)}" + "\n")
40 #print(outliers)
41
42 # PROSENTVIS FEIL
43 df_with_outliers_removed = df_with_outliers_removed.copy()
44 df_with_outliers_removed["ErrorPercent"] = (df_with_outliers_removed["WeightDifference"].abs() / (df_with_outliers_removed["RealWeight"] - CUP_WEIGHT_G)) * 100
45
46 # STATISTIKK
47 statistics = df[["WeightDifference", "TimeSpentMinutes"]].describe()

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47 statistics_outliers_removed = df_with_outliers_removed[["WeightDifference",
48 "TimeSpentSeconds", "ErrorPercent"]].describe()
49
50 # Fjerner koppene med 10% fylling, brukes i Analyse-kapittel
51 df_trimmed_no10percentcups = df_with_outliers_removed.drop(df_with_outliers_re-
52 moved[df_with_outliers_removed["WaterPercentage"] == 10].index)
53
54 # %%
55
56 def get_weight_MAE(df: pd.DataFrame) -> float:
57     return df['WeightDifference'].abs().mean()
58
59 def get_mean_error_percent(df: pd.DataFrame) -> float:
60     return df['ErrorPercent'].mean()
61
62 def plot_weight_error(df: pd.DataFrame):
63     errors = df['WeightDifference'].sort_values().reset_index(drop=True)
64     plt.figure(figsize=(10, 5))
65     plt.bar(errors.index, errors.values)
66     plt.xlabel('Flaske nummer sortert etter netto vektavvik')
67     plt.ylabel('Netto vektavvik [g]')
68     plt.title('Distribusjon av netto vektavvik [g] (sortert stigende)')
69     plt.tight_layout()
70     plt.grid(axis='y')
71     plt.show()
72
73 def plot_time_distribution(df: pd.DataFrame):
74     plt.figure(figsize=(10, 5))
75     plt.hist(df['TimeSpentSeconds'], bins=30, edgecolor='black')
76     plt.xlabel('Syklustid per kopp (sekunder)')
77     plt.ylabel('Antall kopper med denne tiden')
78     plt.title('Distribusjon av syklustid per kopp')
79     plt.xticks(np.arange(26, df['TimeSpentSeconds'].max() + 0.5, step=0.5))
80     plt.yticks(np.arange(0, 30, step=2))
81     plt.tight_layout()
82     plt.grid(axis='y')
83     plt.show()
84
85 def plot_error_percent(df: pd.DataFrame):
86     errors = df['ErrorPercent'].abs().sort_values().reset_index(drop=True)
87     plt.figure(figsize=(10, 5))
88     plt.bar(errors.index, errors.values)
89     plt.xlabel('Flaske (sortert etter vektavvik i %)')
90     plt.ylabel('Netto vektavvik [%]')
91     plt.title('Netto vektavvik som prosent av fyllnivå (sortert stigende)')
92     plt.tight_layout()
93     plt.grid(axis='y')
94     plt.show()
95
96 def plot_error_by_cup(df: pd.DataFrame):
97     fill_levels = {i: f'Kopp {i}\n({i*10}%)' for i in range(1, 11)}
98     avg_error = df.groupby('CupNo')['ErrorPercent'].mean().reindex(range(1, 11))
99     plt.figure(figsize=(10, 5))
100    plt.bar(avg_error.index, avg_error.values)
101    plt.xlabel('Kopp nummer / Fyllnivå')
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99     plt.ylabel('Gjennomsnittlig netto avvik (MAPE) [%]')
100     plt.title('Gjennomsnittlig netto avvik (MAPE) % for hver kopp nummer')
101     plt.xticks(range(1, 11), [fill_levels[i] for i in range(1, 11)])
102     plt.tight_layout()
103     plt.grid(axis='y')
104     plt.show()
105
106 def plot_abs_error_by_cup(df: pd.DataFrame):
107     fill_levels = {i: f'Kopp {i}\n({i*10}%)' for i in range(1, 11)}
108     avg_error = df.groupby('CupNo')['WeightDifference'].apply(lambda x:
109 x.mean()).reindex(range(1, 11))
110     plt.figure(figsize=(10, 5))
111     plt.bar(avg_error.index, avg_error.values)
112     plt.xlabel('Kopp nummer / Fyllnivå [%]')
113     plt.ylabel('Gjennomsnittlig vektavvik per fyllingsgrad [g]')
114     plt.title('Gjennomsnittlig vektavvik per fyllingsgrad [g]')
115     plt.xticks(range(1, 11), [fill_levels[i] for i in range(1, 11)])
116     plt.tight_layout()
117     plt.grid(axis='y')
118     plt.show()
119
120 # KJØR PLOTS
121 plot_weight_error(df_with_outliers_removed)
122 plot_time_distribution(df_with_outliers_removed)
123 plot_error_percent(df_with_outliers_removed)
124 plot_error_by_cup(df_with_outliers_removed)
125 plot_abs_error_by_cup(df_with_outliers_removed)
126
127 # PRINT RESULTATER
128 MAE_weight = get_weight_MAE(df_with_outliers_removed)
129 mean_error_pct = get_mean_error_percent(df_with_outliers_removed)
130 print(f"MAE vekt: {MAE_weight:.2f}g")
131 print(f"MAPE: {mean_error_pct:.2f}%")
132 #print(statistics_outliers_removed)
133 print(get_mean_error_percent(df_trimmed_no10percentcups))
```